

TASK 1

Implement the numerical solution using finite difference method for the following ordinary differential equation and compare with the analytical solution:

$$\begin{aligned}\frac{du}{dx} &= kx, & 0 \leq x \leq 1, \\ \text{At } x = 0, & \quad u = 0, \\ \text{At } x = 1, & \quad \frac{du}{dx} = 1\end{aligned}$$

Objective: The objective of this task is to solve the given ordinary differential equation analytically and numerically using the Finite Difference Method (FDM). The numerical solution is implemented in MATLAB and compared with the analytical solution to verify the accuracy of the method.

The complete solution, including the analytical derivation, finite difference formulation, MATLAB implementation, and results, is provided in the following pages.

Analytical Solution and Finite Difference Formulation:

(Derivation attached below.)

Date ___/___/___



Task 1

method 1 $\frac{dv}{dn} = kn \quad 0 \leq n \leq 1$ at $n=0 \quad v=0$
at $n=1 \quad \frac{dv}{dn} = 1$

— given at $n=1 \quad \frac{dv}{dn} = 1$

$\therefore kn=1 \Rightarrow k=1$

now $\frac{dv}{dn} = n$

$$dv = n dn$$
$$v = \frac{n^2}{2} + C$$

at $n=0 \quad v=0 \Rightarrow C=0$

$$\therefore v = \frac{n^2}{2}$$

method 2 finite difference method

$$\frac{dv}{dn} \approx \frac{v_{i+1} - v_i}{s}$$

s = grid spacing

$$\therefore \frac{v_{i+1} - v_i}{s} = n_i$$

$$v_{i+1} = v_i + s n_i$$

$$v_0 = 0$$

now, the numerical solution is obtained iteratively.

MATLAB Implementation :

```
clc;
clear;
close all;

N = 20;
h = 1/N;

x = linspace(0,1,N+1);

u_exact = 0.5*x.^2;

u_num = zeros(1,N+1);
u_num(1) = 0;

for i = 1:N
    u_num(i+1) = u_num(i) + h*x(i);
end

error = abs(u_exact - u_num);

fprintf(' x\t\tNumerical\tExact\t\tError\n');

    x           Numerical      Exact      Error

for i = 1:length(x)
    fprintf('%0.4f\t\t%0.6f\t%0.6f\t%0.6f\n', ...
    x(i), u_num(i), u_exact(i), error(i));
end

0.0000      0.000000      0.000000      0.000000
0.0500      0.000000      0.001250      0.001250
0.1000      0.002500      0.005000      0.002500
0.1500      0.007500      0.011250      0.003750
0.2000      0.015000      0.020000      0.005000
0.2500      0.025000      0.031250      0.006250
0.3000      0.037500      0.045000      0.007500
0.3500      0.052500      0.061250      0.008750
0.4000      0.070000      0.080000      0.010000
```

0.4500	0.090000	0.101250	0.011250
0.5000	0.112500	0.125000	0.012500
0.5500	0.137500	0.151250	0.013750
0.6000	0.165000	0.180000	0.015000
0.6500	0.195000	0.211250	0.016250
0.7000	0.227500	0.245000	0.017500
0.7500	0.262500	0.281250	0.018750
0.8000	0.300000	0.320000	0.020000
0.8500	0.340000	0.361250	0.021250
0.9000	0.382500	0.405000	0.022500
0.9500	0.427500	0.451250	0.023750
1.0000	0.475000	0.500000	0.025000

```

figure;
plot(x,u_exact,'LineWidth',2);
hold on;
plot(x,u_num,'o--','LineWidth',1.5);
grid on;
xlabel('x');
ylabel('u(x)');
title('Analytical vs Numerical Solution');
legend('Analytical','FDM');
saveas(gcf,'comparison_plot.png')

```

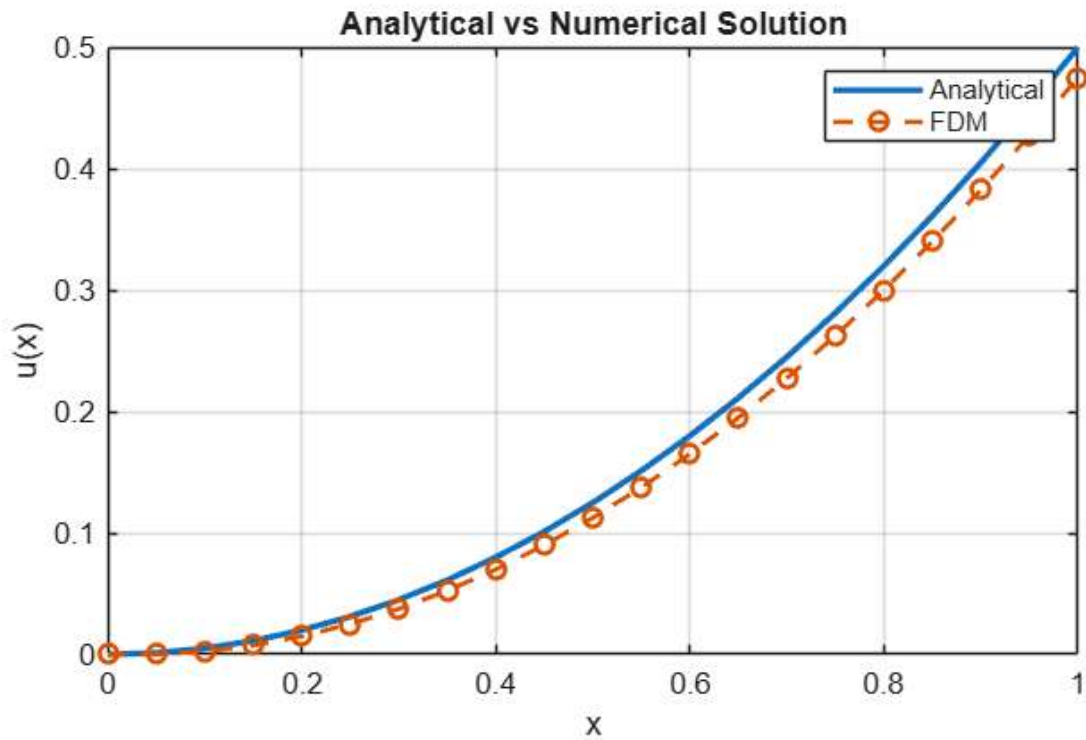


Figure 1: Comparison of Analytical and Numerical Solutions

```
figure;
plot(x,error,'o-','LineWidth',1.5);
grid on;
xlabel('x');
ylabel('Absolute Error');
title('Error Distribution');
saveas(gcf,'error_plot.png')
```

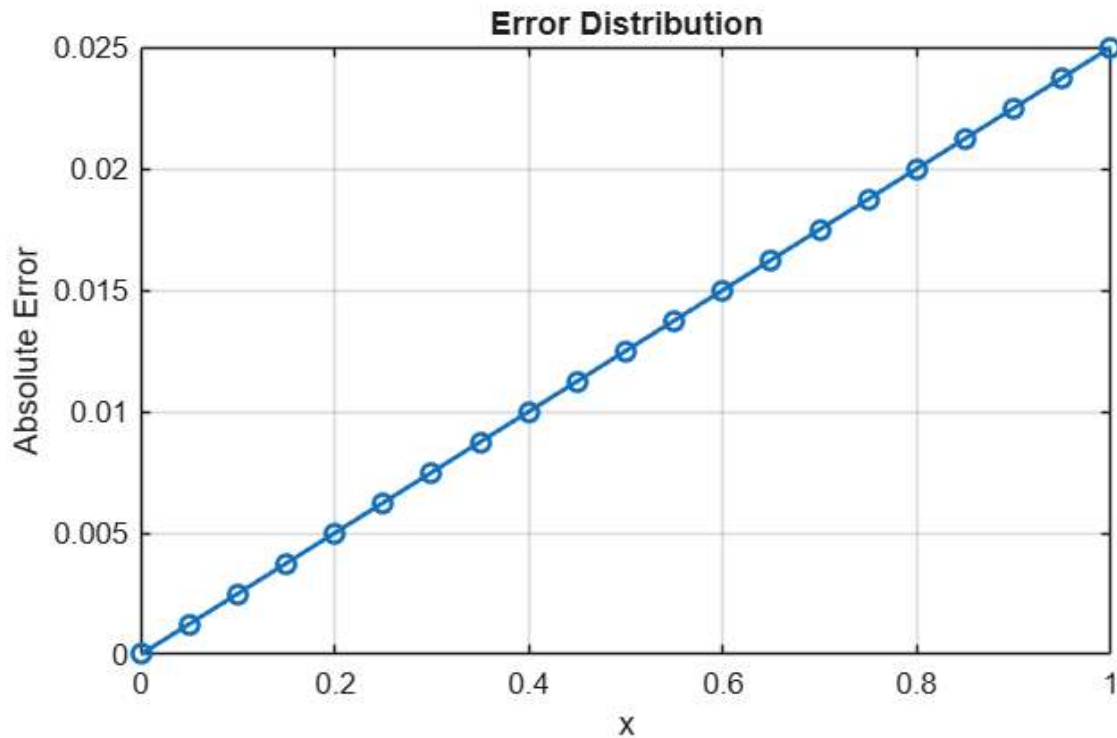


Figure 2: Absolute Error Distribution

```
fprintf('\nMaximum Error = %e\n', max(error));  
Maximum Error = 2.500000e-02
```

The given differential equation was solved analytically and numerically using the Finite Difference Method (FDM). The MATLAB implementation produced results that closely matched the analytical solution, demonstrating the validity of the numerical approach.

GitHub Repository: <https://github.com/mugdhajha/Project-Research.git>

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